YUNJAE LEE

Seoul National University ◇ Seoul, Republic of Korea
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RESEARCH INTERESTS

High-speed mixed-signal and RF ICs, wireline and optical transceiver circuits, clock and frequency generation, and analog-to-digital converters

EDUCATION

Seoul National University (SNU) Seoul, Republic of Korea
B.S. Candidate in Electrical and Computer Engineering Mar. 2021 – Present

- Cumulative GPA: 3.94/4.30
- Relevant Coursework: Analog Integrated Circuits (A+), Microwave Engineering (A), Design Project for Electrical Devices & Systems (A+)
- Mandatory military service: Mar. 2022 – Sep. 2023

Incheon Academy of Science and Arts (IASA) Incheon, Republic of Korea
High School for Gifted Students in STEM Mar. 2018 – Jan. 2021

RESEARCH OUTPUTS

Manuscripts Under Review

1. **Yunjae Lee**, Minchul Kim, Uichan Park, Taeyeong Yoon, and Jungsuek Oh. "Equivalent-Circuit-Assisted Design of a Screen-Printed Wideband Metasurface Absorber." Under review at *IEEE Access*.

Patents

1. Jaehoon Shin, **Yunjae Lee**, Yuncheol Shin, Wonjun Kang, and Daegi Cho. "Pedestrian Safety System on the Crosswalk." Republic of Korea Patent No. 10-2263981, granted Jun. 7, 2021.

RESEARCH EXPERIENCE

Integrated Circuits and Systems for Computing and Communication Lab (IC3) Seoul, Republic of Korea
Jan. 2026 – Present
Research Intern – Advisor: Prof. Wooseok Choi

- Conducting research on mixed-signal ICs, focusing on high-speed interfaces and phase-locked loops.
- Designed a hybrid PLL for a B.S. thesis, combining a linear proportional path with digital integral and double-integral control based on a published architecture.
- Implemented oscillator-current-dependent biasing in the proportional path to scale loop bandwidth with reference frequency, while incorporating coarse and fine frequency-control paths to address the trade-off between oscillator tuning range and frequency resolution.

Wave Fusion Lab (WFL) Seoul, Republic of Korea
Research Intern – Advisor: Prof. Jungsuek Oh Jan. 2025 – Dec. 2025

- Developed a full-wave-calibrated equivalent circuit model to guide resonance tuning and impedance matching in a screen-printed wideband metasurface absorber.
- Participated in the industry-collaborative research project *Efficient Design of RF Components and Systems Based on Artificial Intelligence* with Samsung Electronics through the Future Talent Internship Program.
- Experimentally characterized the absorption performance of AI-designed metasurface patterns.
- Analyzed ray-tracing models incorporating diffuse scattering.

SELECTED PROJECTS

Dynamic Vision Sensor Pixel Design Jul. 2026 – Present
AI Semiconductor Analog Circuit Design Competition

- Designing a 45-nm CMOS DVS pixel in Cadence Virtuoso, targeting low noise, a low contrast threshold, and in-pixel event storage.
- Developing a dedicated circuit block to suppress charge injection and mitigate unintended pixel behavior.

Wideband Amplifier Design Oct. 2025 – Dec. 2025
Course Project, Microwave Engineering

- Designed and optimized a wideband amplifier using Advanced Design System (ADS), focusing on precise component selection and impedance matching.

FPGA Digital Clock Dec. 2024
Course Project, Logic Design and Laboratory
- Co-developed a multifunction digital clock in Verilog on a Basys 3 FPGA using Xilinx Vivado, integrating timekeeping, alarm settings, a stopwatch, and an LED-based alarm-dismissal game.
- Integrated modular timing and control logic with button debouncing, input synchronization, and multiplexed seven-segment display control.

TECHNICAL SKILLS

Programming Python, MATLAB, Verilog, C, C++, LaTeX
Tools Cadence Virtuoso, ADS, Ansys HFSS, Xilinx Vivado, Vector Network Analyzer, KiCad

HONORS AND AWARDS

Selected as an IPESK Next-Generation Engineer Sep. 2026
Institute for Promotion of Engineering and Science of Korea (IPESK)

Future Talent Internship Scholarship Seoul, Republic of Korea
Department of Electrical and Computer Engineering Apr. 2025 – Dec. 2025
Seoul National University

Silver Award, Creative Design Fair Seoul, Republic of Korea
College of Engineering, Seoul National University Sep. 2024
Team Project: AI-Based Smart IP Camera

National Scholarship for Science and Engineering 2021, 2024 – 2026
Korea Student Aid Foundation (KOSAF)
Four-year full-tuition scholarship awarded for outstanding academic performance upon admission.

Korea Talent Award Dec. 2020
Awarded by the Minister of Education, Republic of Korea.

Bronze Award, Samsung Humantech Paper Award Feb. 2019
Samsung Electronics
Awarded for a paper on crosswalk lighting design to reduce nighttime vehicle-pedestrian collisions.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant - Analog Electronic Circuits Seoul, Republic of Korea
Department of Electrical and Computer Engineering, Seoul National University Sep. 2026 – Dec. 2026
- Support students by answering questions and clarifying concepts in analog electronic circuits during and outside of class.

Laboratory Teaching Assistant - Basic Circuit Theory and Laboratory Seoul, Republic of Korea
Department of Electrical and Computer Engineering, Seoul National University Mar. 2024 – Jun. 2024
- Guided students in using laboratory equipment and assisted with circuit experiments and troubleshooting.

Guest Instructor 2023 – 2024
Data Analysis with Python
- Delivered one-day introductory lectures on data analysis using Python to high school students at multiple schools.

EXTRACURRICULAR ACTIVITIES & VOLUNTEERING

Volunteer, Youth Volunteer Group Seoul, Republic of Korea
Haeyul Law Firm Apr. 2026 – Present

Team Leader / Member, *Young-Sang* Seoul, Republic of Korea
SNU Photography Club, Seoul National University Mar. 2021 – Dec. 2024
Served as a team leader in 2024, coordinating group photography projects.